



Development of a Low-Power Alpha Wearable Prototype for Ear-Based Temperature Monitoring to prevent Heat Injuries in Industrial Settings

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Introduction

Industrial workers often operate in high-temperature environments while performing intense physical tasks. Prolonged exposure increases the risk of heat-related illnesses such as heat exhaustion and heat stroke.

OSHA (2023) reports ~30,000 cases and ~100 deaths annually in the U.S. due to occupational heat stress. Current body temperature monitoring systems are often bulky, expensive, and incompatible with required PPE, making them impractical on job sites.

Objective

Develop a compact, low-power, ear-based temperature monitoring system that:

- Integrates with PPE (noise cancellation earplugs, hardhats)
- Supports real-time Core Body Temperature tracking using an in-ear thermistor
- Provides auditory and haptic alerts to the worker
- Enables data access via USB-C and BLE for safety monitoring

This alpha-stage prototype aims to improve heat stress management in industrial environments.

Methods

Prototyping began with schematic and PCB design using EasyEDA and KiCad. SolidWorks was used to model the enclosure and in-ear interface, followed by 3D printing and silicone molding. Firmware was developed in CircuitPython and Arduino for BLE communication, data logging, and auditory/haptic alerts. A custom Python–PyQt6 desktop app enabled multi-device monitoring. Initial sensor data was validated using a handheld laser thermometer, with further testing scheduled.

System Architecture and Design

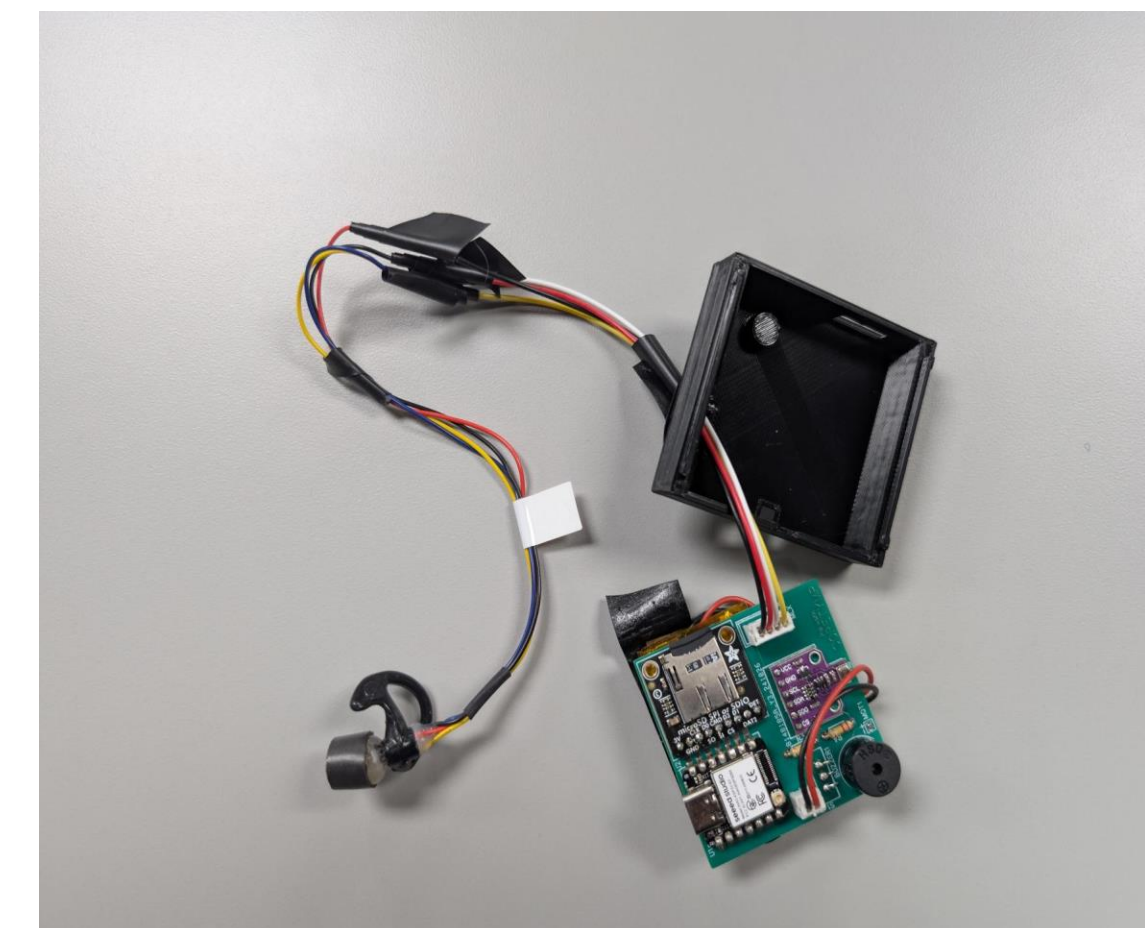


Fig: Initial PCB Assembly
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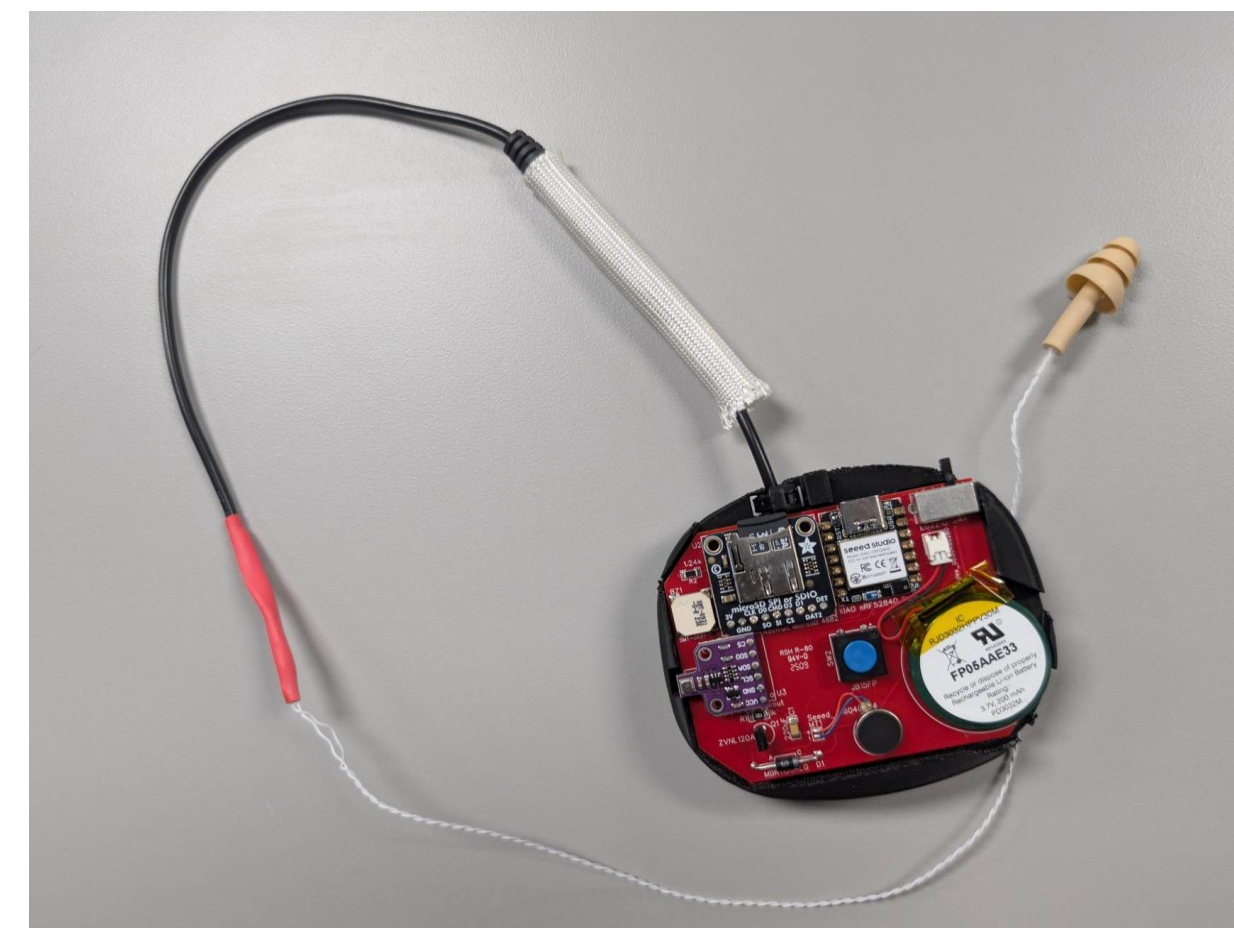
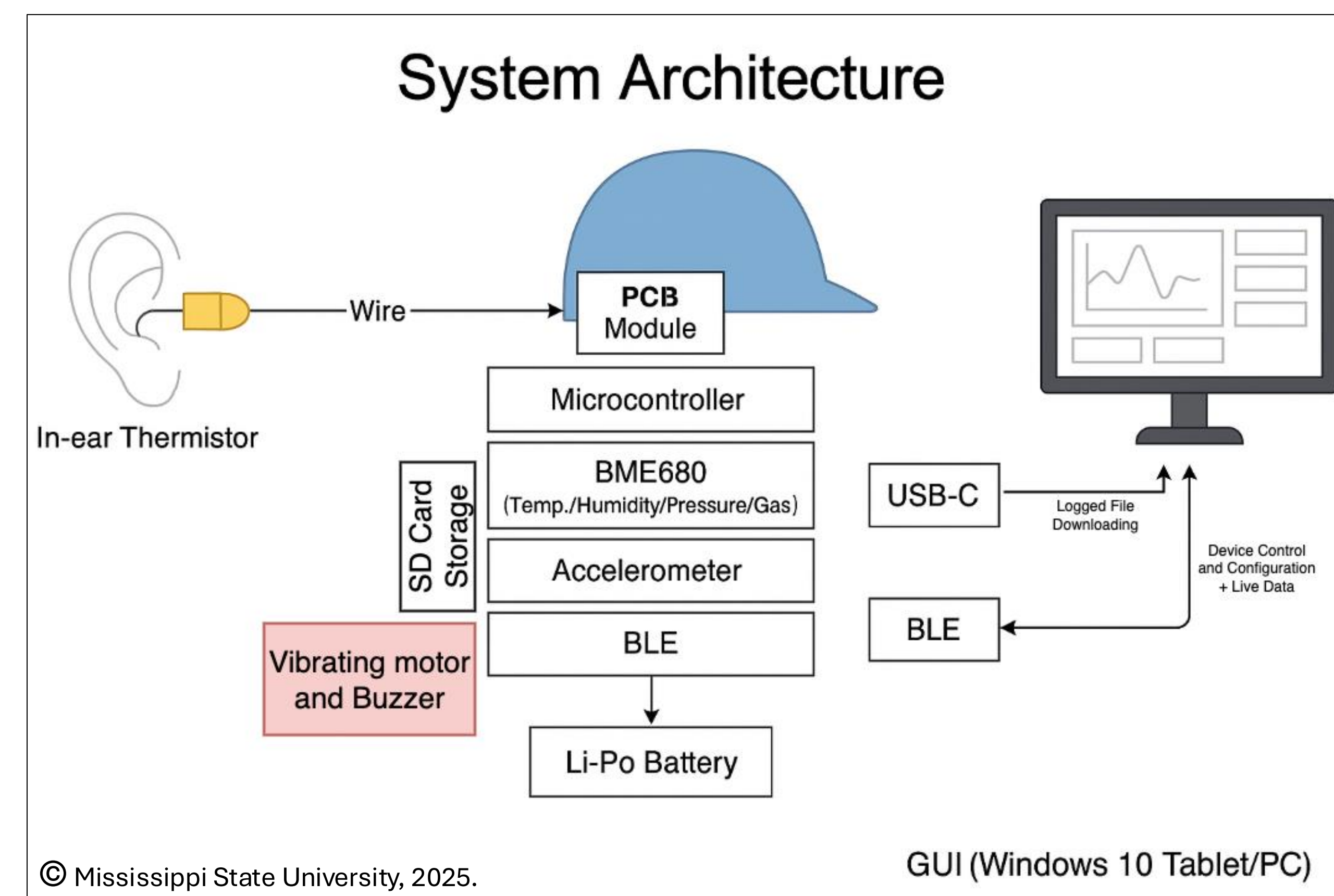


Fig: Updated PCB Assembly
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Fig: Hardhat Assembly
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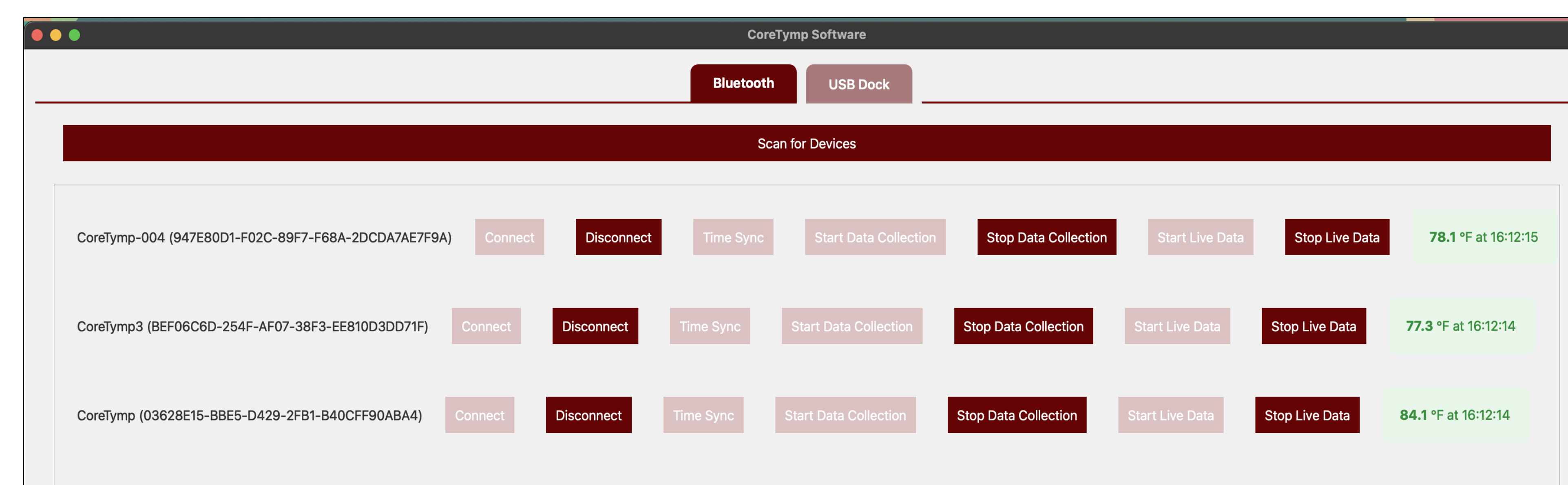


Fig: GUI Dashboard, © Mississippi State University, 2025.

Device Capabilities

- Power and Connectivity
 - ~16 mA average, 12 hours continuous use
 - BLE for live data, control, config
 - USB-C with custom UART for data export
- Sensing Frequency
 - Core temp (ear): 10/min
 - Env. (BME680): temp, humidity, pressure, gas every min
 - Acceleration: 2 Hz continuous
- Alerts
 - Buzzer/beeper, configurable thresholds (eg: 95°F watch, 98°F warning, 101°F critical)
- Data Logging and Config
 - SD card, 100+ hours logging capacity
 - Download via GUI or USB-C
 - JSON file for default settings (sampling, thresholds, etc.)
- Form Factor
 - Integrated with earplug and hardhat
 - Weight ~80 g

Validation and Testing

- Lab testing against core temperature pills for accuracy
- Field deployment of 10 prototypes at a shipbuilding company
- Collect user feedback and usage data
- Use findings to inform next hardware and software revision

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